16.1 Heuristics for Crossing Number

We were wondering, in the introduction to the chapter, if there is anything we can do to get us out of this tight spot with crossing number analysis; the truth is that, to the best of current knowledge, we can't do much, at least unless we relax the requirement for a deterministic algorithm guaranteeing the correct answer and return an embedding with a graph's *minimum crossing number* (or *minimum rectilinear*[3] *crossing number*).

A common strategy to deal with *NP-hard* problems, in fact, is to use heuristics: these algorithms, often operating non-deterministically, are able to provide a sub-optimal answer in a reasonable amount of time.

We have already met randomized algorithms throughout this book, for instance in chapters 3 and 4, but if you feel you can use a refresher, feel free to take a look at appendix F, which is providing a brief summary of randomized algorithms.

16.1.1 Did You Just Say Heuristics?

I realize the concept of heuristic might be confusing: why would we be fine with an algorithm that doesn't return the correct answer? Sometimes, indeed, we can't: there are problems were we absolutely need to get the right answer, even if we need to wait more – like, for instance, if you are running a nuclear power plant or designing a new drug, you don't want to settle on a sub-